

Foundation (Jalapeno)

Q1. Simplify $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. Simplify $\sqrt[3]{27}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. Convert $\sqrt{x}$ to rational exponent form

- (A) $x^{1/2}$
- (B) $x^{1/3}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q4. Simplify $\sqrt{64}$

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 14

Q5. A spaceship travels $\sqrt{121}$ kilometers in 2 hours. How many

kilometers does it travel per hour?

- (A) 5 km/h
- (B) 6 km/h
- (C) 7 km/h
- (D) 8 km/h
- (E) 11 km/h

Q6. Simplify $\sqrt[3]{64}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q7. Convert $x^{1/3}$ to radical form

- (A) $\sqrt[2]{x}$
- (B) $\sqrt[3]{x}$
- (C) $\sqrt[4]{x}$
- (D) $\sqrt[5]{x}$
- (E) $\sqrt[6]{x}$

Q8. A lab experiment requires $\sqrt{25}$ grams of a substance. How many grams are needed?

- (A) 3 grams
- (B) 5 grams
- (C) 7 grams
- (D) 9 grams
- (E) 11 grams

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{48}$. Show all work.

Q10. Convert $x^{2/3}$ to radical form and simplify $\sqrt[3]{128}$. Show all work.

Chef's Tips

- Read the questions carefully
- Use a ruler to draw diagrams
- Check your work